

VAIBHAV KULKARNI

Backend Software Engineer

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PROFESSIONAL SUMMARY

Backend-focused Software Engineer with 4 years of experience building scalable microservices, workflow automation platforms, distributed systems, and cloud-native applications. Strong in Node.js, NestJS, and TypeScript with PostgreSQL, RabbitMQ, and Kafka, and hands-on with Kubernetes, Docker, and Angular. Owns systems end-to-end from database schema and REST API design through deployment, with proven production ownership — resolving high-severity incidents, optimizing performance, and delivering platform engineering and SDK infrastructure adopted across teams.

TECHNICAL SKILLS

- **Backend:** Node.js, NestJS, Express.js, TypeScript, REST APIs
- **Frontend:** Angular, JavaScript, HTML, CSS, RxJS
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, TypeORM, Query Optimization
- **Messaging:** RabbitMQ, Kafka, Event-Driven Architecture
- **Cloud & DevOps:** Docker, Kubernetes, Helm, Azure, AWS, CI/CD, Nx Monorepo
- **Architecture:** Microservices, Distributed Systems, System Design, SDK Design, Multi-Tenancy
- **Other:** jBPM, Workflow Automation, Production Support, Performance Optimization, C++

PROFESSIONAL EXPERIENCE

Neutrinos MXP LLP — Software Engineer (SDE II)

Aug 2022 - Present | Bengaluru

Alpha — Low-Code Workflow Automation Platform (NestJS, Angular, RabbitMQ, PostgreSQL, jBPM, Kubernetes, Helm)

- Built config-type adapters and implemented JSON diffing with tag-based config fetching on top of the platform's Change Tracking (Config Ledger) system, and resolved defects across config versioning flows.
- Built and published reusable SDKs and shared libraries (config/API and admin-filters) consumed across multiple backend services, reducing duplicated configuration code.
- Engineered a production BPM Archival service for backup, restore, and deletion of high-volume process data with restartable batch processing, keeping production databases performant at scale; hardened it with an end-to-end test suite.
- Delivered a RabbitMQ-backed task-distribution system end-to-end (entities, service, REST API, audit logging) and migrated the integration to quorum queues, improving message durability and failover.
- Resolved the three highest-severity production incidents of the year (production outage, session-expiry crash, task re-attempt blocker), owning root-cause analysis and recovery for customer-facing environments.
- Optimized a critical API 95% (200ms to 10ms) via PostgreSQL query tuning and stabilized a production database, cutting CPU from 90%+ spikes to a steady ~50%.
- Drove the Helm chart migration ahead of a major release and standardized Kubernetes deployment manifests and Dockerfiles across 14+ services; built the platform's database migration and audit pipeline for safe, traceable schema evolution.
- Acted as integration and release owner, reconciling release, hotfix, and snapshot branches to keep the platform continuously shippable.

Neutrinos Studio — Low-Code Full-Stack Builder (Angular, Node.js, Electron)

- Resolved 30+ production issues in 30 days, owning platform stability during a critical hardening window and reducing customer-reported failures.
- Built a database entity designer and internal tooling that accelerated customer-application generation for internal builders.

PROJECTS

- **Vibes Shield — OIDC Identity Platform:** Production-grade authentication and authorization system implementing JWT, RBAC, and OIDC on a secure NestJS + PostgreSQL backend, delivering centralized, standards-based identity for downstream services.
- **Vibes DS Server:** Redis-like in-memory datastore built with a custom C++ HTTP server and low-level socket networking, demonstrating systems-level performance engineering and protocol handling.
- **Chat Service:** Real-time WebSocket messaging service on Node.js deployed to AWS EC2, supporting low-latency bidirectional communication.

EDUCATION

Bachelor of Engineering (BE), The National Institute of Engineering, Mysore | Aug 2018 - Jul 2022